

Sense — Second Proof of Concept

Overview

The second proof of concept was designed to upgrade the first prototype, moving from static, wired tests to a real-time wireless setup.

This version focuses on measuring resistance changes across conductive elements within the sensing layer, enabling the detection of fluid absorption over time while streaming live data via Bluetooth.

The same physical sensor setup and wiring structure were used as in POC #1, but the system was upgraded to an Arduino Nano 33 BLE Sense Rev 2 to support real-time BLE transmission.

System Overview

The measurement system is based on a modified version of the Arduino capacitance meter described in the following guide:

 [Measure Capacitance with Arduino – Instructables](#)

This system allows the estimation of low capacitance values in the picofarad (pF) range, suitable for detecting small changes in the tampon's capacitance due to fluid absorption.

Hardware Setup

Component-	Description-
Arduino Nano 33 BLE Sense Rev 2	Upgraded controller with built-in Bluetooth Low Energy, used to send resistance readings in real time.
Sensor Layer	Two conductive zinc-strip electrodes embedded in a tampon analog; resistance decreases as liquid bridges the electrodes.
Circuit Layout	Identical to POC #1 - voltage divider configuration with fixed capacitor and sensor element as variable resistor.
Power Supply	USB or external 5 V supply.
Testing Medium	Saline “synthetic blood” solution applied incrementally to simulate progressive absorption.

Firmware & Communication

- The firmware continuously reads the analog voltage across the sensor's resistor pair and converts it into a corresponding resistance value.
- Readings are transmitted as BLE characteristic updates using the Arduino Nano 33 BLE Sense Rev 2's built-in Bluetooth module.
- Data was observed and visualized in our prototype mobile application, which displayed live fill-level readings in real time.

Results & Observations

- Real-time **wireless resistance readings** successfully achieved.
- Sensor output clearly responded to increased liquid presence, confirming reliable detection.
- Resistance values correlated with absorption rate in synthetic medium, demonstrating strong repeatability between tests.

